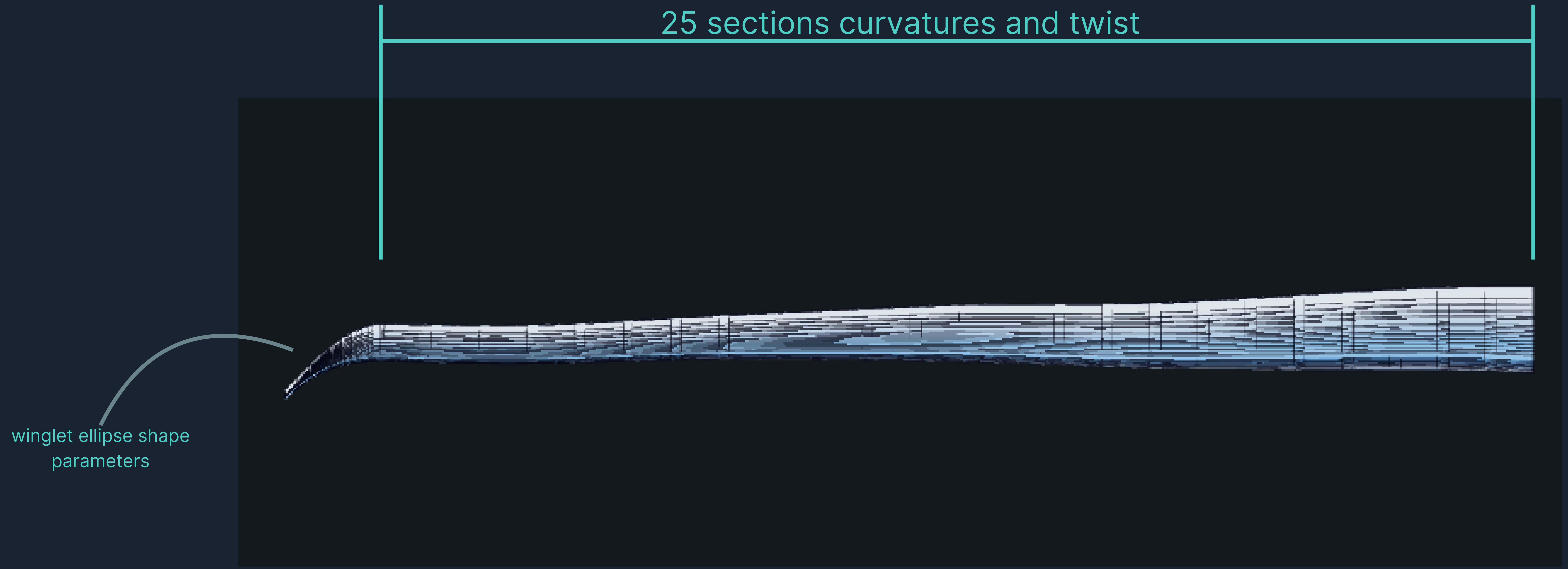
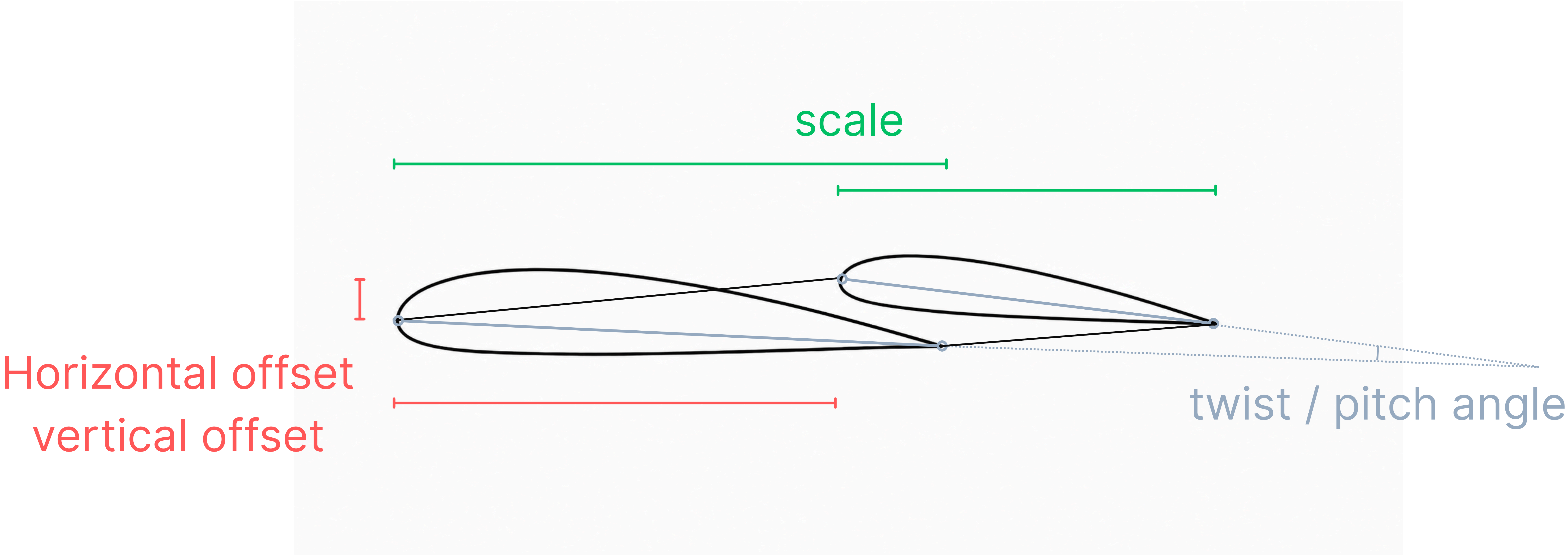
Genetics Aerodynamics Optimisation

Wing Design Through Evolutionary Algorithms

Parameters


25 sections curvatures and twist
winglet ellipse shape parameters

Spanwise Sections Parameters

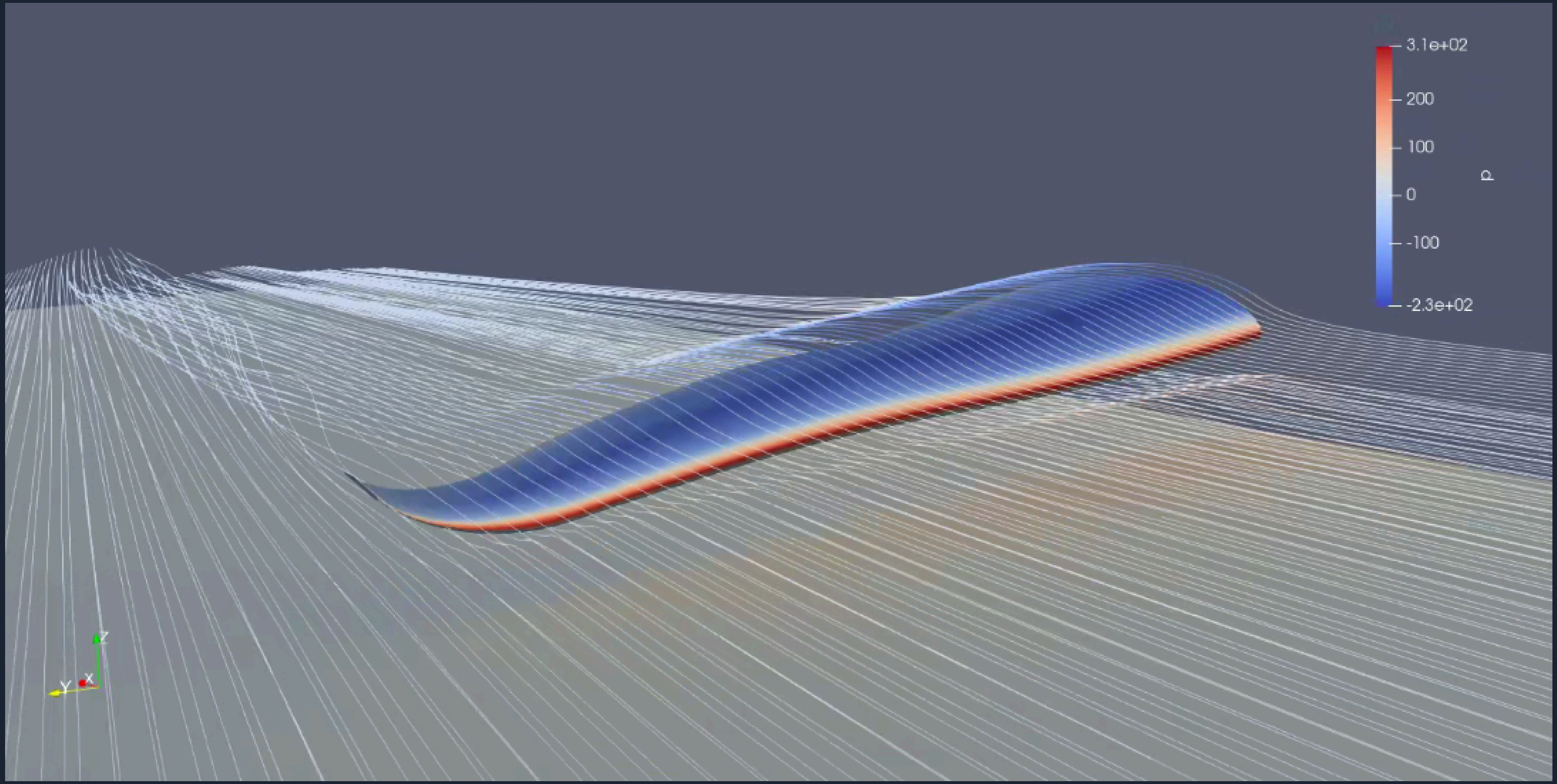


Winglet Parameters

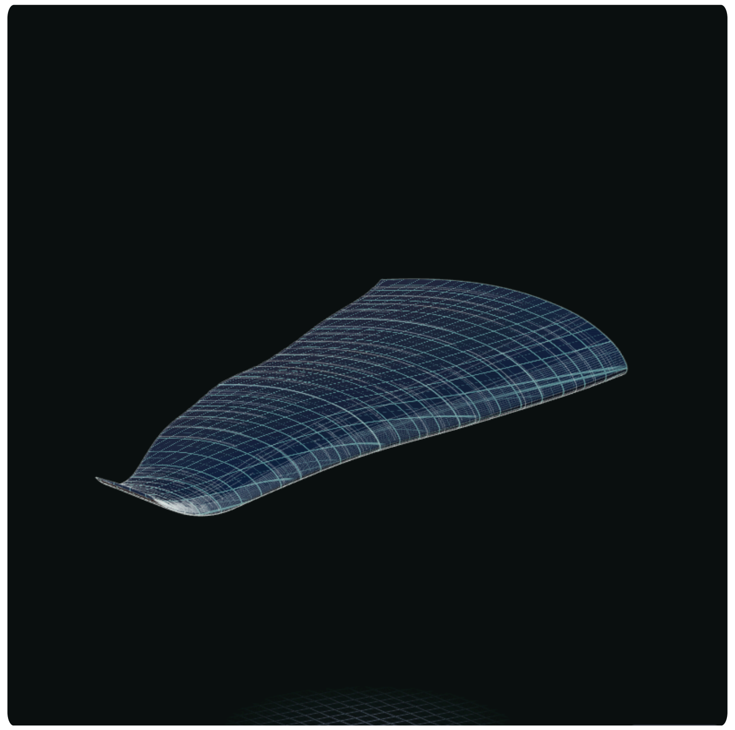
pointy/ not pointy end
ellipse length
ellipse radius
ellipse arc angle
rotations
scale
horizontal offset

Genesis Batch
Seed Generation

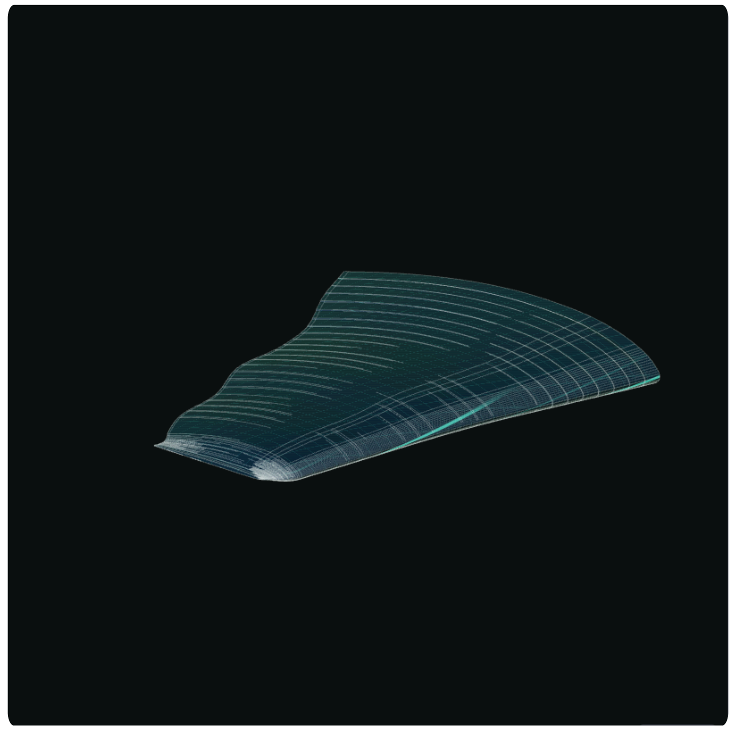
OpenFOAM
CFD



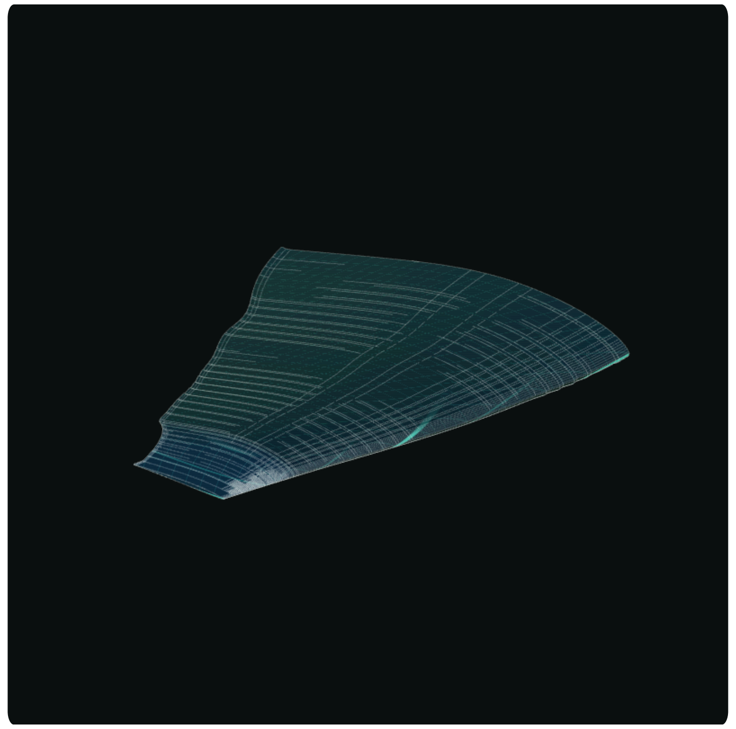
Genesis Batch - Survivors



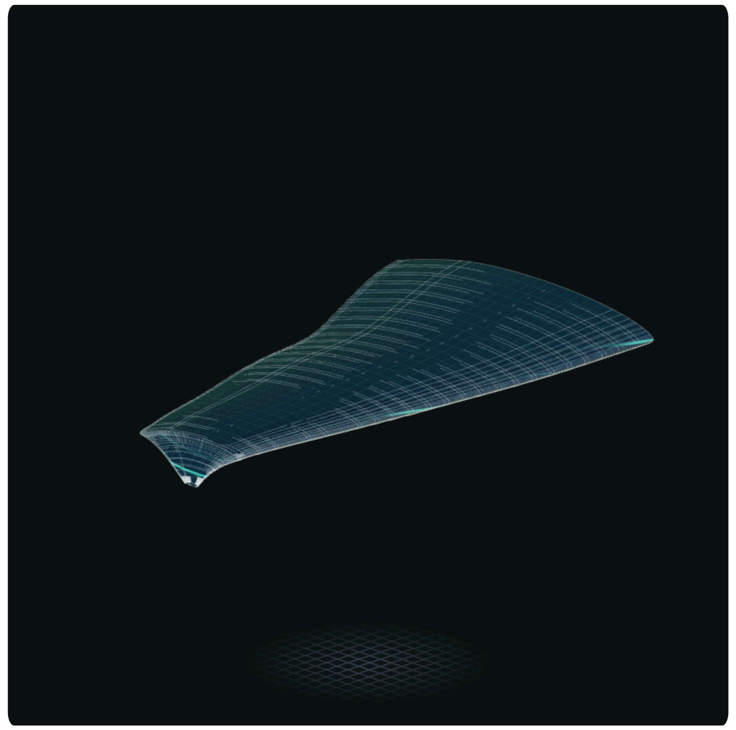
design_000



design_005



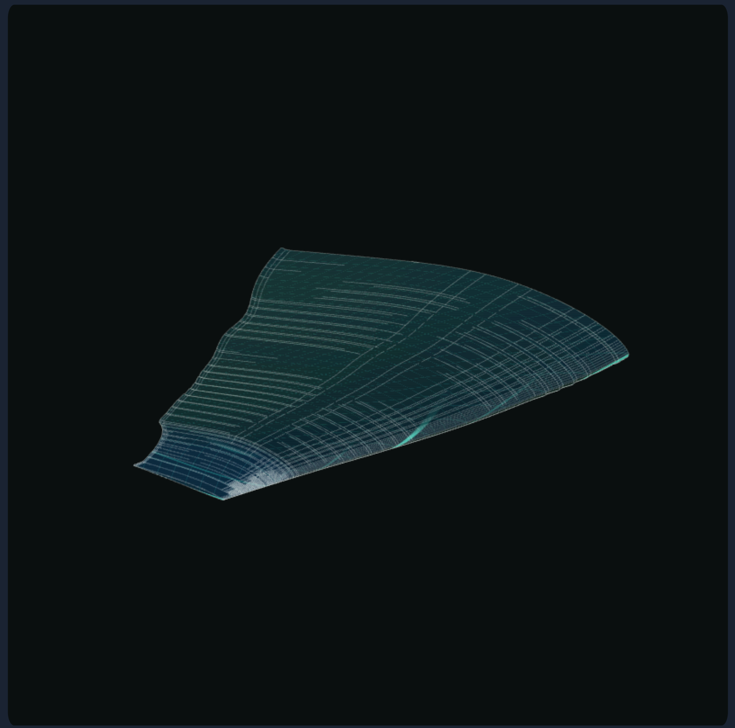
design_007



design_004

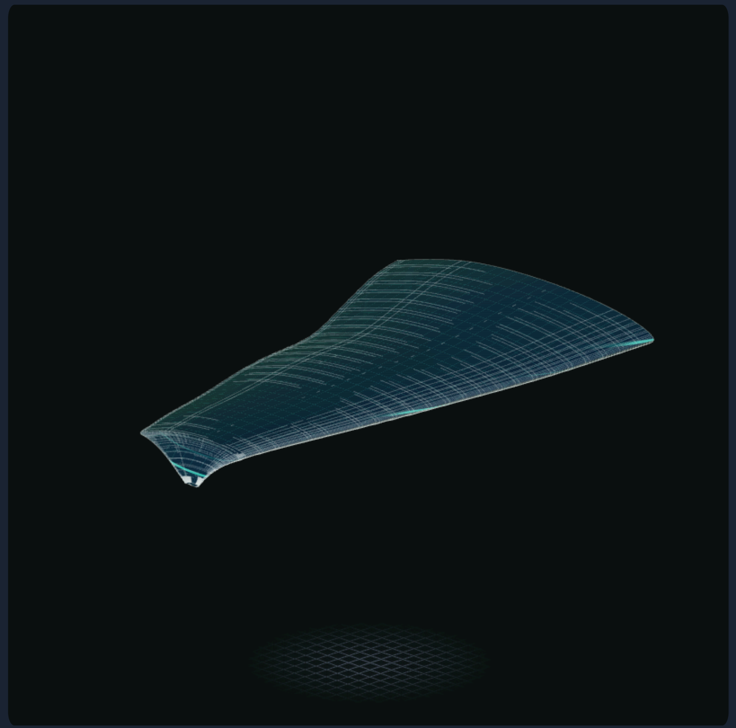
design_id	rank	role	score	eligible	reason	cl	cd	lift_to_drag	ground_effect_g	abs_cm	span_per_ld
design_000	1	best	1.52286	TRUE		0.354715	0.0179515	19.9623	9.88817	0.0115698	0.088519
design_005	2	best	1.256919	TRUE		0.297035	0.0174404	17.0813	4.39544	0.00691968	0.0943422
design_007	3	runner_up	1.205106	TRUE		0.20452	0.0150709	13.5938	3.6973	0.00348772	0.120036
design_004	4	runner_up	1	TRUE		0.545357	0.0232517	24.9097	27.3918	0.135068	0.0653907
design_003	5		0.948693	TRUE		0.430054	0.0215877	20.5762	17.1311	0.0814697	0.0746982
design_008	6		-1.012674	TRUE		0.0108222	0.0149127	0.743768	-3.22996	0.0348071	2.17723
design_001	7		-1.664398	TRUE		-0.0126135	0.0139172	-0.673802	-12.6147	0.0909094	inf
design_002				FALSE	only 4/5 cases						
design_006				FALSE	only 1/5 cases						
design_009				FALSE	only 0/5 cases						

Gen - 01
Breeding Next Gen

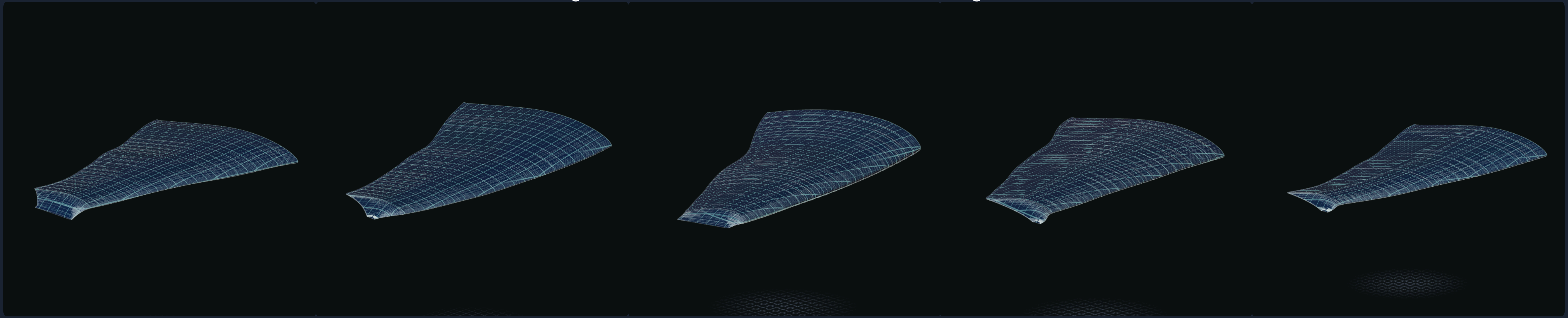


design_007

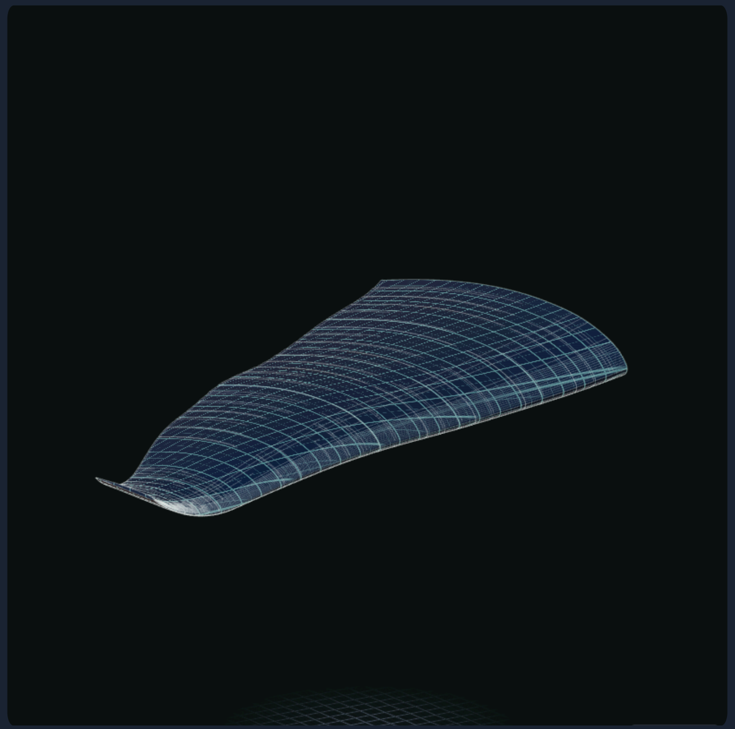
X



design_004

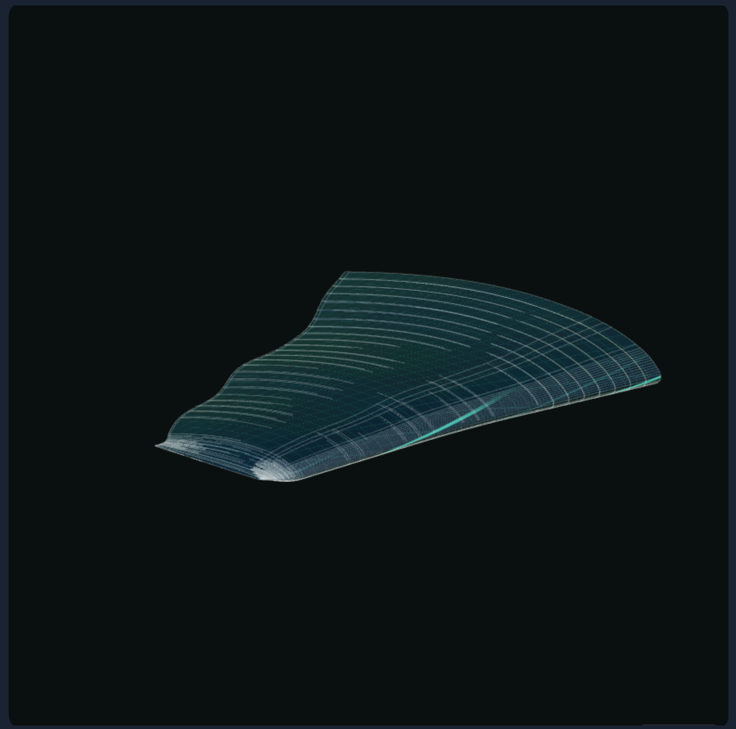


Gen - 01
Breeding Next Gen

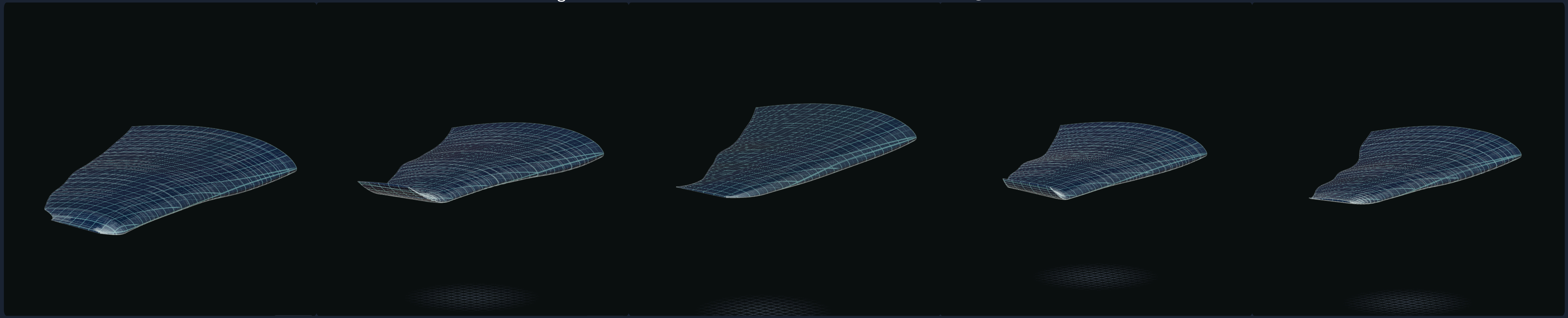


design_000

X

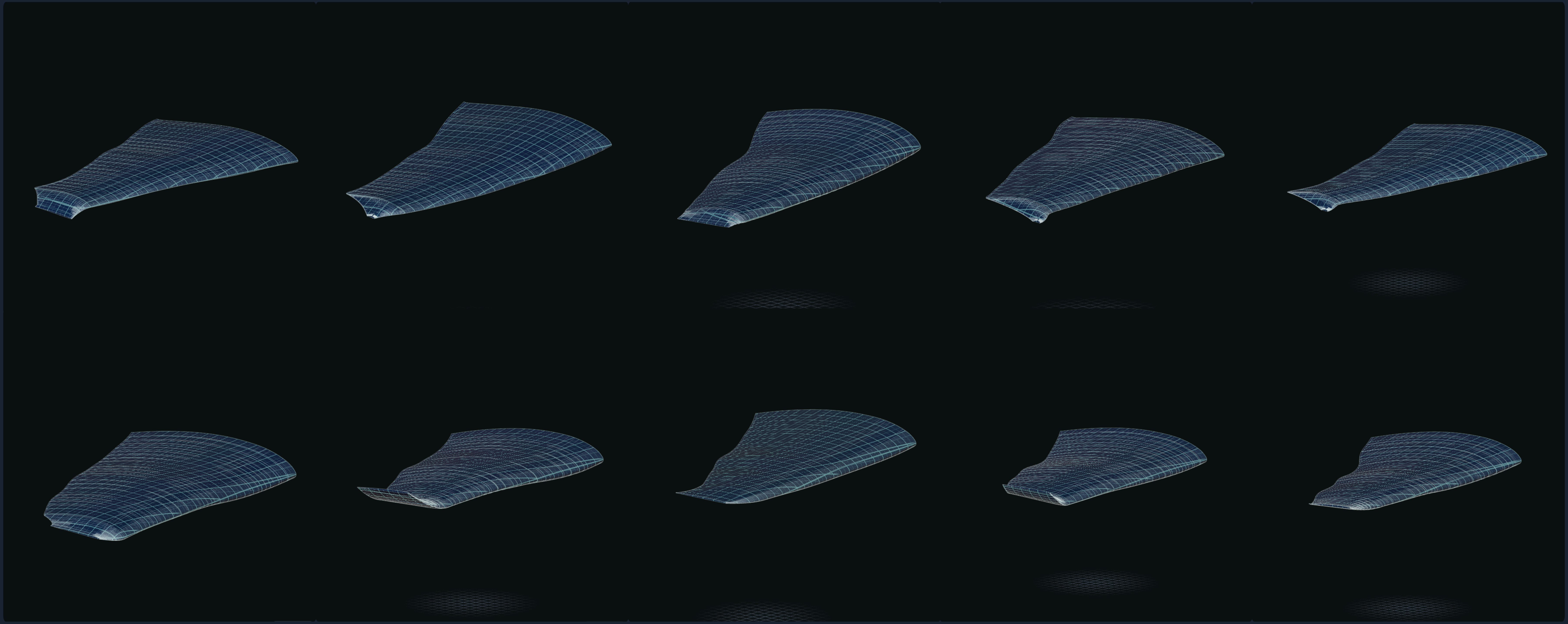


design_005



Gen - 01

Breeding Next Gen



Gen - 01
Breeding Log


```json
  },
  "fitness": {
    "_comment": "Weighted sum of min-max normalised metrics (each scaled 0..1 across eligible designs). Positive weight = higher metric is better. cl/cd/lift_to_drag/abs_cm are means over the valid height sweep; ground_effect_gain = -d(L/D)/d(height) from a least-squares fit (steeper L/D rise as the ground gets closer = bigger = better); span_per_ld = full span incl. winglet reach divided by mean L/D (compactness). *_at_lowest_height metrics are extra knobs, weight 0 by default.",
    "weights": {
      "cl": 1.0,
      "cd": -1.7,
      "lift_to_drag": 1.3,
      "ground_effect_gain": 1.0,
      "abs_cm": -0.7,
      "span_per_ld": -0.8,
      "ld_at_lowest_height": 0.2,
      "cl_at_lowest_height": 0.2
    },
    "evaluation_height_m": null
  },
  "selection": {
    "num_best": 2,
    "num_runner_ups": 2,
    "runner_up_strategy": "next_best",
    "_runner_up_strategy_options": "next_best = ranks 3 and 4; diverse = from the next diverse_pool_size ranked designs, pick the 2 genetically furthest from the already-selected survivors",
    "diverse_pool_size": 5
  },
  "breeding": {
    "children_per_pair": 5,
    "pairing": "random_disjoint",
    "crossover_units": {
      "airfoil": "whole",
      "wing_pitch": "per_section",
      "wing_horizontal": "per_section",
      "wing_vertical": "per_section",
      "wing_scale": "per_section",
      "winglet": "per_gene"
    },
    "_crossover_units_options": "whole = the group is inherited intact from parent A or B (50/50); per_gene = every gene in the group is drawn from A or B independently (e.g. each winglet parameter separately); per_section = per-section array genes are mixed element-by-element (each section's slope/scale/pitch-change from A or B; slope arrays are smoothness-repaired afterwards), scalar genes in the group drawn independently"
  },
  "mutation": {
    "rate": 0.10,
    "scale": 0.20,
    "_comment": "rate = per-gene probability (array genes: per element). scale = gaussian sigma as a fraction of each gene's configured range, clamped to bounds. Discrete genes (airfoil, pointy flag) re-draw with the same rate."
  },
```

```json
    "children": [
      {
        "design_id": "design_012",
        "parents": [
          "design_004",
          "design_007"
        ],
        "crossover": {
          "airfoil": "A",
          "wing_pitch": {
            "root_pitch_deg": "A",
            "pitch_change_deg": "ABBAAAABBBAABABAAAAAAAAABA"
          },
          "wing_horizontal": {
            "horizontal_slope": "BAAABABABBBBBAAABBABABAAB"
          },
          "wing_vertical": {
            "vertical_slope": "ABABABBBABABBAABBBAABABBA"
          },
          "wing_scale": {
            "scale_down": "AABAAAABBBBAAAAABBBBABBAA"
          },
          "winglet": {
            "winglet_is_pointy": "B",
            "winglet_ellipse_length": "B",
            "winglet_ellipse_radius": "A",
            "winglet_ellipse_rotation_deg": "B",
            "winglet_arc_angle_deg": "A",
            "winglet_horizontal_rate": "B",
            "winglet_scale_rate": "A",
            "winglet_flat_tip_chord_m": "A",
            "winglet_pointy_tip_x_frac": "A"
          }
        },
        "mutations": [
          "root_pitch_deg: -1.24331 -> -2",
          "winglet_ellipse_length: 0.140699 -> 0.172664",
          "winglet_arc_angle_deg: 41.7324 -> 33.0491",
          "horizontal_slope[10]: -0.280583 -> -0.10783",
          "horizontal_slope[18]: -0.217912 -> -0.163632",
          "vertical_slope[13]: -0.040731 -> 0.0726088",
          "vertical_slope[21]: -0.040731 -> -0.0649213",
          "pitch_change_deg[3]: -0.0956833 -> 0.157731",
          "scale_down[5]: 0.0325072 -> 0.0397229",
          "scale_down[23]: 0.0409651 -> 0.05",
          "root_airfoil: Eppler817 -> ClarkY"
        ],
```